

(12) **Patent Application Publication**
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(10) **Pub. No.: US 2023/0181470 A1**
(43) **Pub. Date: Jun. 15, 2023**

(60) Provisional application No. 62/738,333, filed on Sep. 28, 2018.

Publication Classification

(51) **Int. Cl.**
A61K 9/16 (2006.01)
A61K 31/439 (2006.01)
A61K 31/454 (2006.01)
A61K 9/48 (2006.01)
A61K 9/00 (2006.01)

(52) **U.S. Cl.**
CPC *A61K 9/1652* (2013.01); *A61K 31/439*
(2013.01); *A61K 31/454* (2013.01); *A61K*
9/1611 (2013.01); *A61K 9/485* (2013.01);
A61K 9/4858 (2013.01); *A61K 9/0053*
(2013.01); *A61K 9/4866* (2013.01)

(22) Filed: **Aug. 29, 2022**

Related U.S. Application Data

(63) Continuation of application No. 17/649,826, filed on Feb. 3, 2022, now Pat. No. 11,452,692, which is a continuation of application No. 17/143,904, filed on Jan. 7, 2021, now Pat. No. 11,471,413, which is a continuation of application No. 16/585,532, filed on Sep. 27, 2019, now Pat. No. 10,933,020.

(57) **ABSTRACT**

Provided herein is an oral pharmaceutical composition, comprising a plurality of xanomeline beads having a core comprising xanomeline or a salt thereof; and a plurality of tropium beads having a core comprising a salt of tropium.

COREALIS

Pharma

Stability protocol for Xanomeline / Trospium HCl Pellets

Manufacturing date: 2016DE12, Incubation date: 2016DE16

Project	Lot (Strength)	Packaging	Stability conditions			Time points (month)					
			25°C/ 60% RH	30°C/ 65% RH	40°C/ 75% RH	0	1	2	3	6	EXTRA
286-02	Xanomeline 50mg (lb) Trospium HCl 20mg	30cc HDPE bottles with sealed PP cap	6 bottles (20 capsules/bottle)			2016DE16	2017JA16	2017FE16	2017MR16	2017JN16	TBD
				6 bottles (20 capsules/bottle)			2017JA16	2017FE16	2017MR16	2017JN16	TBD
					4 bottles (20 capsules/bottle)		2017JA16	2017FE16	2017MR16	2017JN16	TBD
	5 bottles (20 capsules/bottle)				2017JA16		2017FE16	2017MR16	2017JN16	TBD	
			6 bottles (20 capsules/bottle)		2017JA16		2017FE16	2017MR16	2017JN16	TBD	
				3 bottles (20 capsules/bottle)	2017JA16		2017FE16	2017MR16	2017JN16	TBD	
Sample to be analyzed for: Appearance, Assay, Related Substances, KF and Dissolution. At the Client request, sample will be removed from chamber and analyzed.											

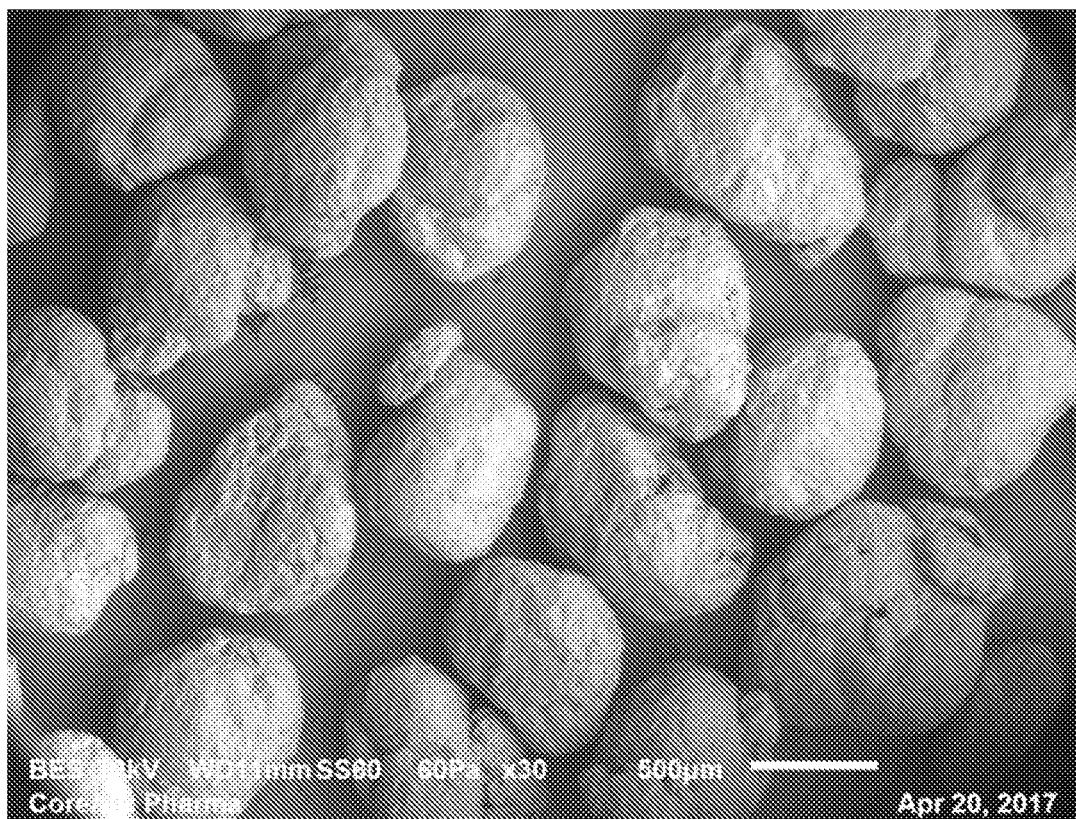


FIG 2



FIG. 3

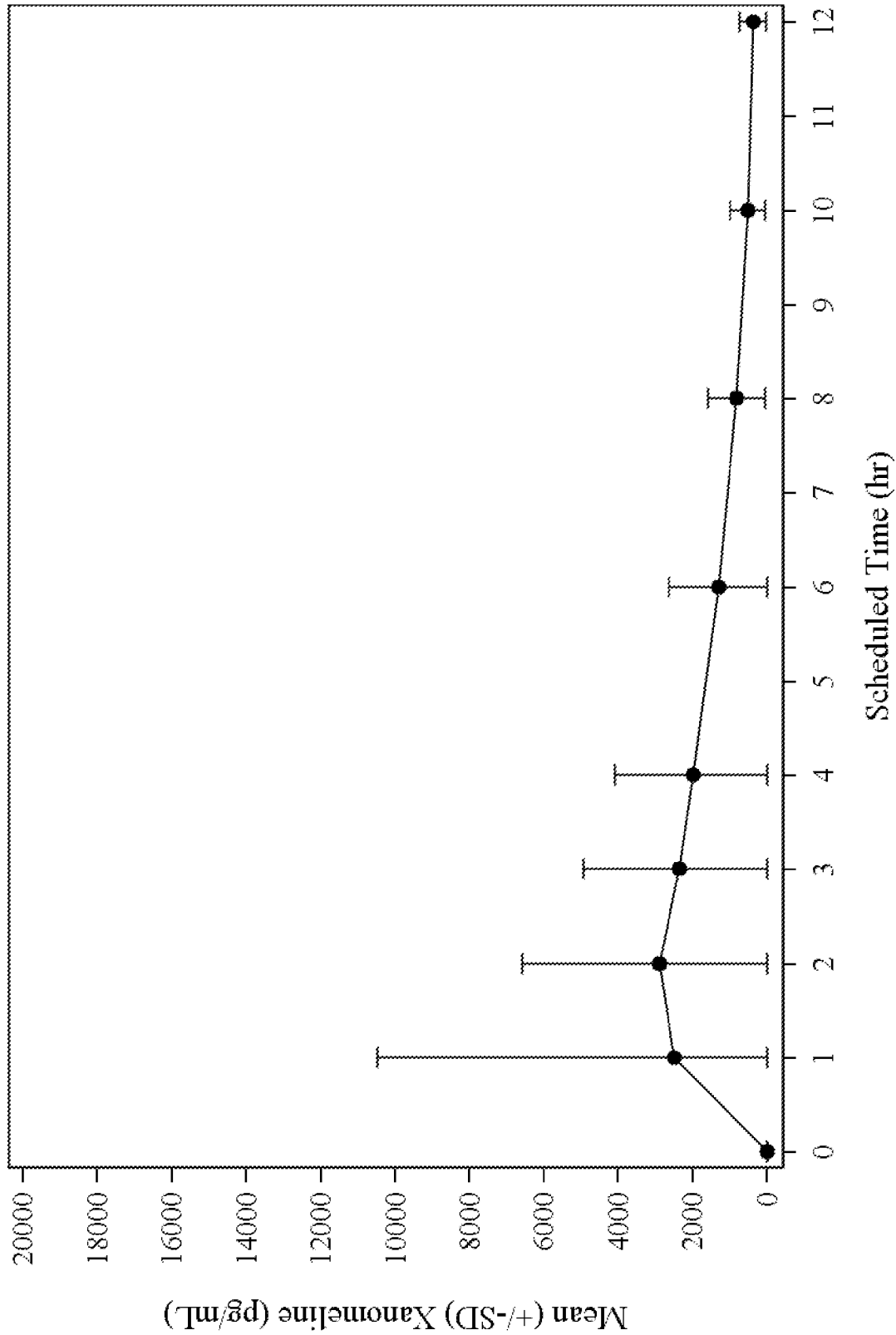


FIG. 42

TABLE 16

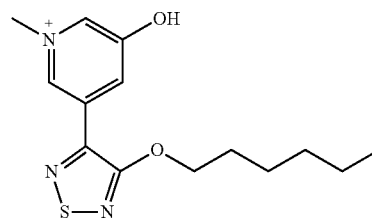
Dissolution of KarXT 50/10						
		Dose strength				
		10 mg Trospium Chloride				
		50 mg Xanomeline free base				
		Active	Trospium Chloride		Xanomeline free base	
		Time (min)	% LC	Range	% LC	Range
Dissolution 900 ml 0.1N HCl Paddles @50 rpm ramp @ 200 rpm after 45 min (n = 3)	T = 0	10	84	85, 86, 82	89	88, 90, 88
		20	96	97, 96, 94	97	96, 96, 98
		30	96	97, 97, 94	97	96, 97, 98
		45	96	97, 96, 94	97	96, 96, 98
		60	96	97, 97, 94	97	96, 96, 98
		(ramp)				
	T = 1 m	10	88	83, 91, 89	88	87, 92, 85
	40° C./	20	101	100, 101, 101	95	96, 97, 94
	75% RH	30	101	101, 101, 101	96	97, 97, 94
		45	101	102, 101, 101	96	97, 97, 94
		60	101	102, 101, 102	96	97, 97, 94
		(ramp)				
	T = 2 m	10	88	89, 91, 83	93	94, 91, 93
	40° C./	20	98	97, 102, 96	99	99, 98, 101
	75% RH	30	99	98, 103, 97	99	99, 98, 101
		45	99	97, 103, 96	99	99, 98, 101
	T = 3 m	10	88	79, 91, 94	93	86, 94, 99
	25° C./	20	99	95, 99, 102	98	95, 97, 102
	60% RH	30	99	95, 99, 102	98	95, 96, 102
		45	99	95, 99, 102	98	95, 96, 102
	T = 3 m	10	90	89, 90, 91	92	90, 95, 90
	40° C./	20	98	99, 95, 99	95	95, 97, 94
	75% RH	30	98	99, 95, 99	95	95, 97, 94
		45	98	99, 95, 99	95	95, 97, 94

[0231] Subsequent testing showed that KarXT 50/10, 50/20, and 75/20 in hard-shell capsules were stable for at least 12 months 25° C./60% RH. Based on available data, a shelf-life of 15 months at 25° C./60% RH is proposed.

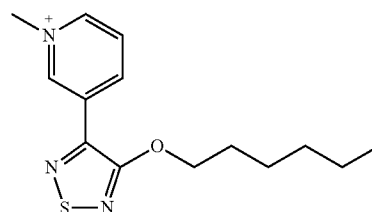
[0232] The dissolution results show that the two compounds release quickly, which may increase their bioavailability, and that they also release at comparable rates despite substantial differences in compositions between the two bead formulations. Both xanomeline and trospium chloride have low bioavailabilities, and rapid release can increase bioavailability by overwhelming saturable processes that limit absorption into the general circulation.

[0233] An unknown xanomeline impurity with a relative retention time of about 1.09 was observed during stability studies of the combination drug products. The impurity was first observed during testing at the three-month time point for the 50 mg xanomeline/10 mg trospium chloride drug product and at the initial time point for the other three combination products, both of which occurred at the same time. The impurity peak increased both with time and with increasing storage temperature. The impurity had not been observed before the present studies.

[0234] Preliminary studies suggest that the RRT 1.09 impurity is 3-[(4-hexyloxy)-1,2,5-thiadiazol-3-yl]-5-hydroxyl-1-methylpyridin-1-ium ($C_{14}H_{20}N_3O_2S^+$, MW=294.1271 Da):



[0235] The RRT 1.09 impurity is a hydroxylated version of Compound V ($C_{14}H_{20}N_3OS^+$, MW=278.1322 Da), which is the penultimate intermediate in the synthesis of xanomeline with negative mutagenic potential:



[0236] To reduce the presence of the impurity, the storage temperature for the drug product was lowered. Bottles were